



布魯科技有限公司

Bulu Tech co., Ltd

MaxGauge For Oracle

Product Introduction

I N D E X



1. 產品概要

- MaxGauge 架構



2. 主要功能

- Real-Time Monitor
- Performance Analyzer



01

產品概要

MaxGauge For Oracle Overview

- MaxGauge 架構





MaxGauge Overview

“為資料庫性能最優化提供的專業化軟體解決方案”

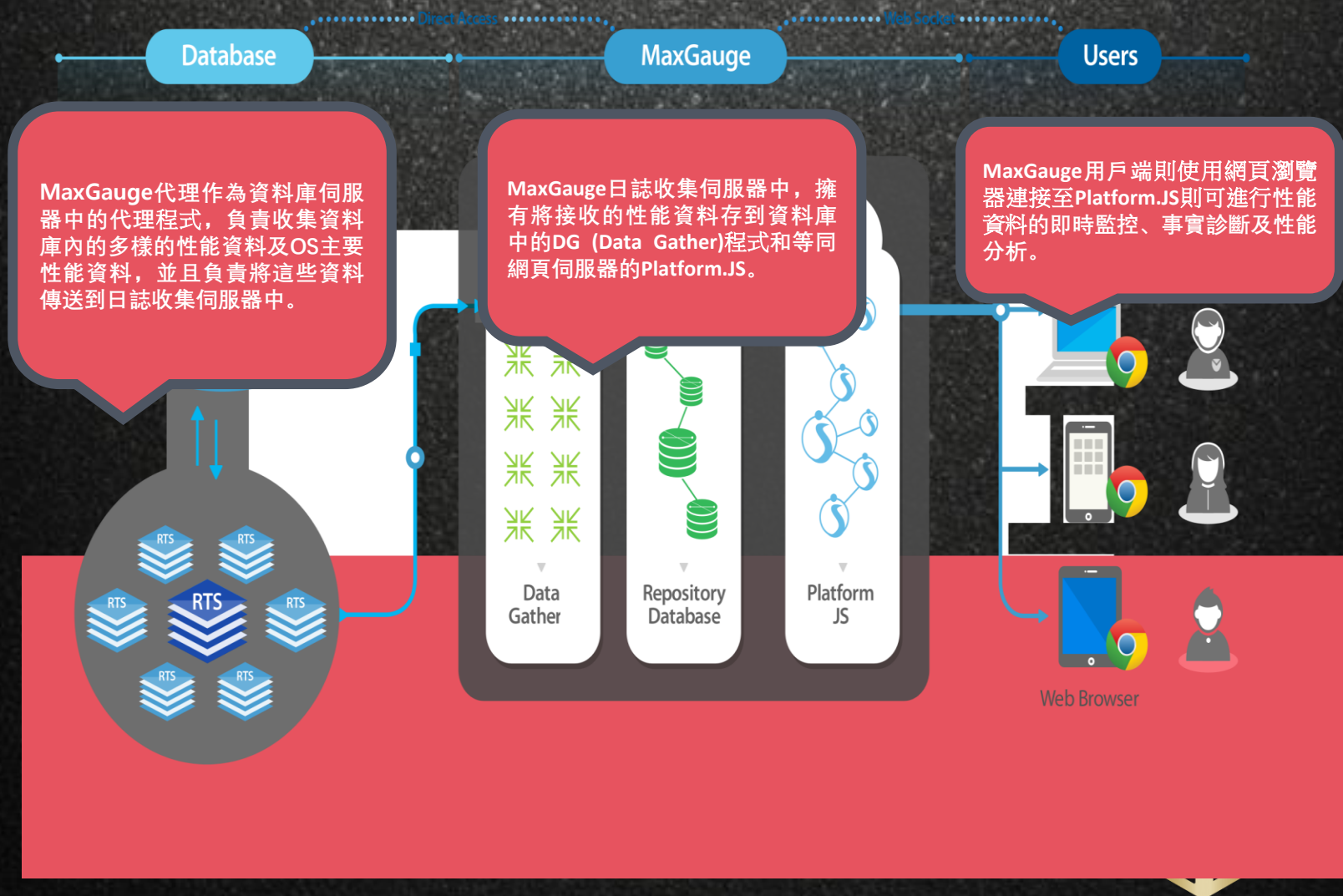
MaxGauge為資料庫性能管理的專業化軟體解決方案，提供實時監控、工作資訊收集、精密診斷及分析、SQL調優等多樣的功能，以此有效管理資料庫系統的可用性及性能。通過提供主動的監控手段，可以快速定位問題情況，有效分析應用程式的瓶頸現象，存儲多樣的性能資料，從而可以為投資效益比提供充足資料。





MaxGauge Architecture

“所有性能資料以**最低負載**進行收集與儲存”





MaxGauge Comparative Advantages

“利用代理的Direct Memory Access方式 收集所有性能資料”

發生複雜性能問題時
無法進行監控及分析

資料庫crash時，
也可以進行監控及精密分析

- 從 SQLPlus login 資料庫，對 v\$, x\$ 進行 query

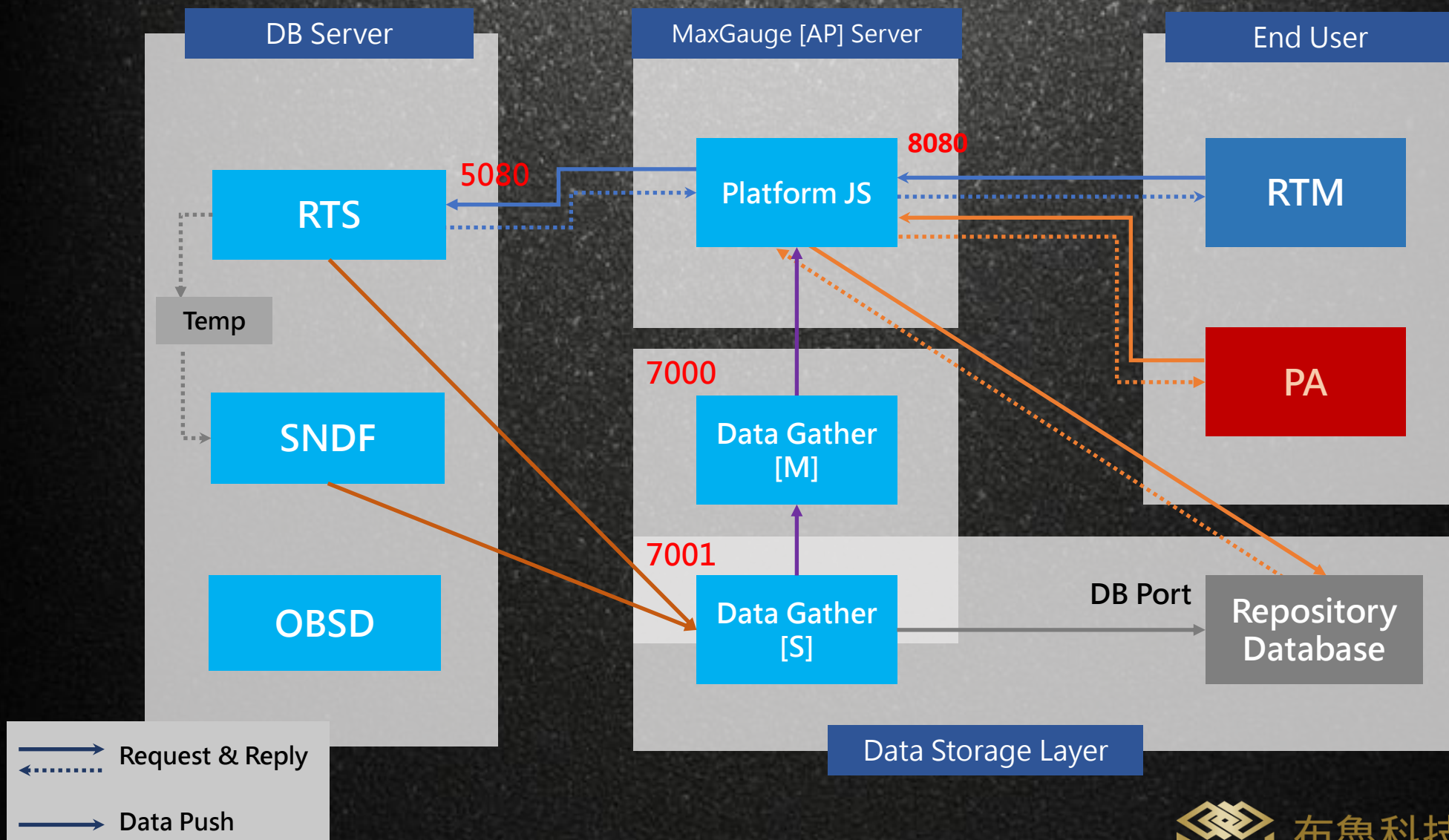


- 從 OS Level 對 SGA 所在記憶體位置進行採樣





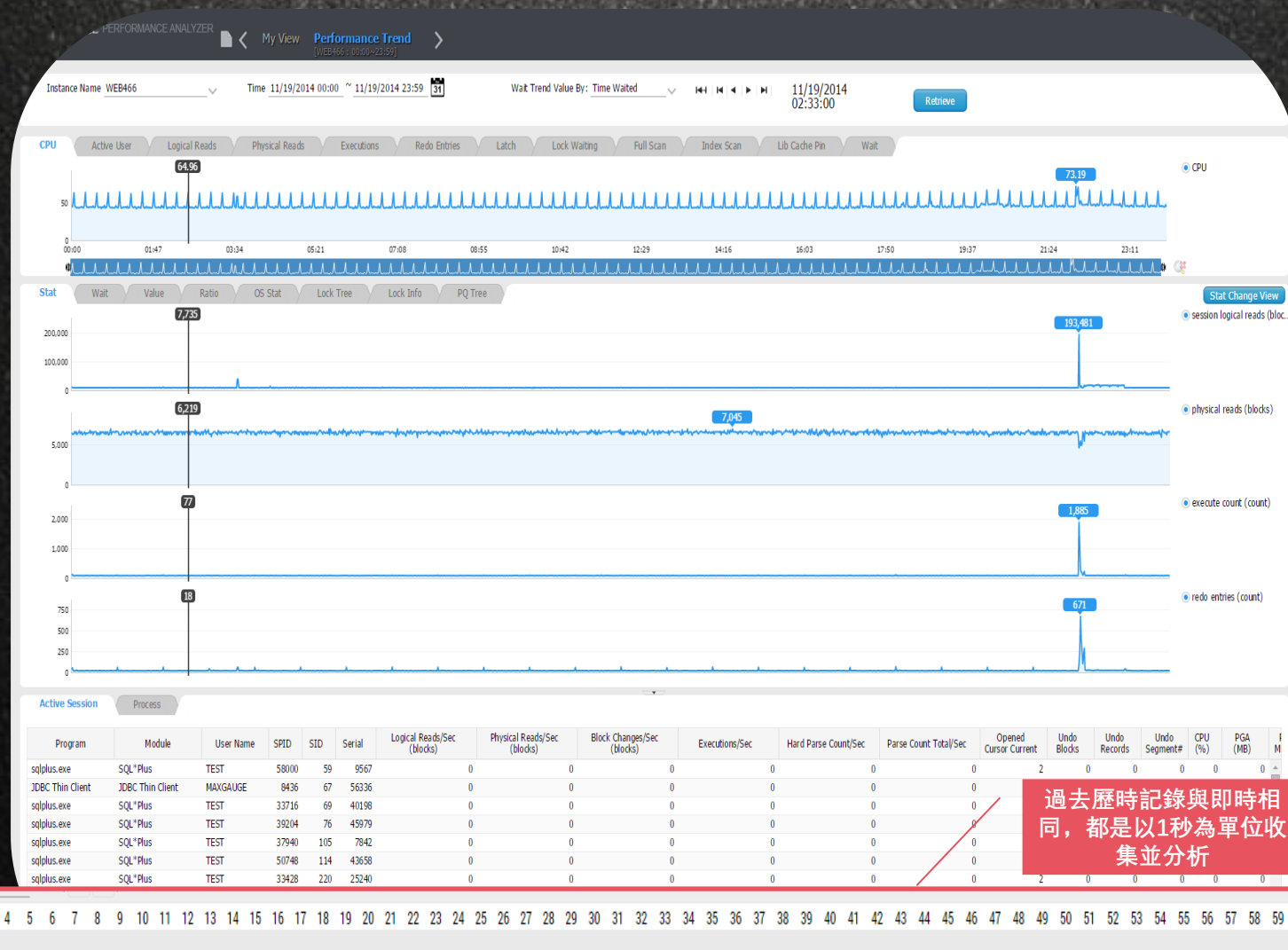
MaxGauge Architecture – Data Flow





MaxGauge Comparative Advantages

“收集所有的工作歷史記錄為1秒單位的資料存儲”

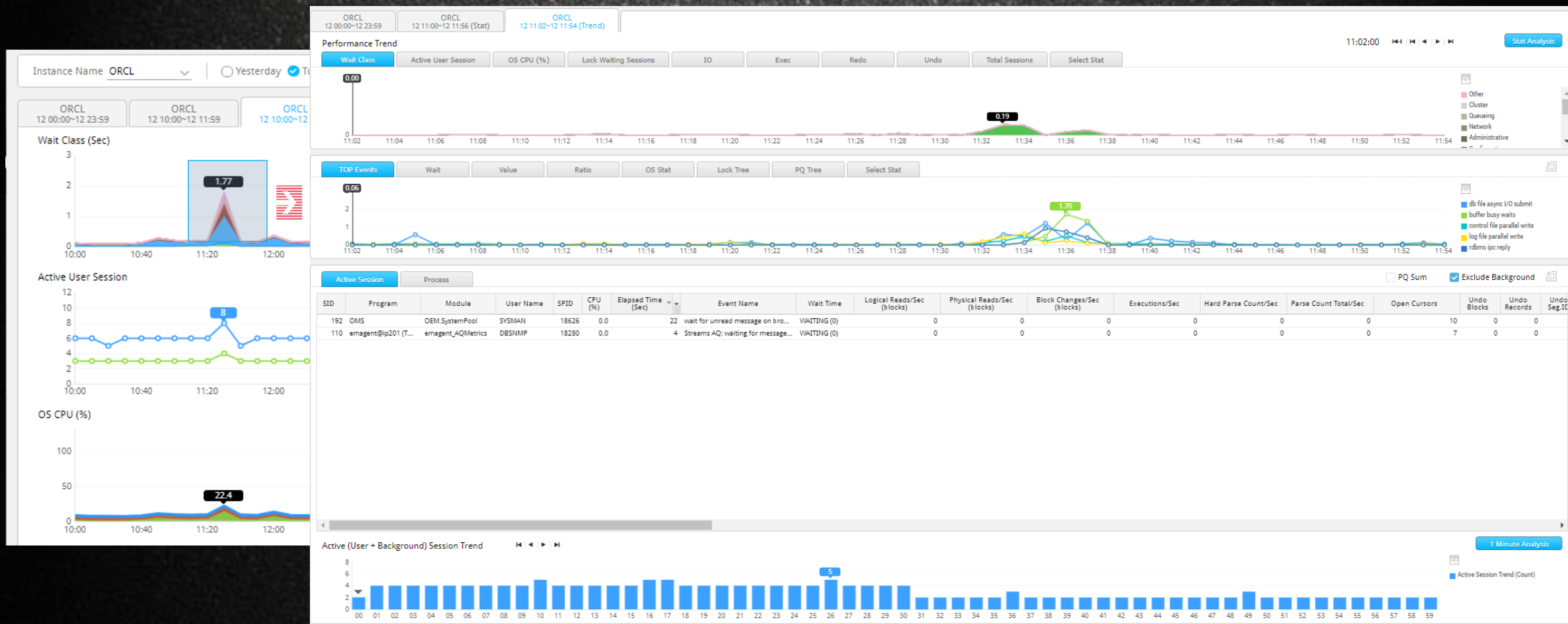


過去歷時記錄與即時相同，都是以1秒為單位收集並分析



MaxGauge Comparative Advantages

“直觀和簡單的使用者介面”





MaxGauge Comparative Advantages



02

主要功能

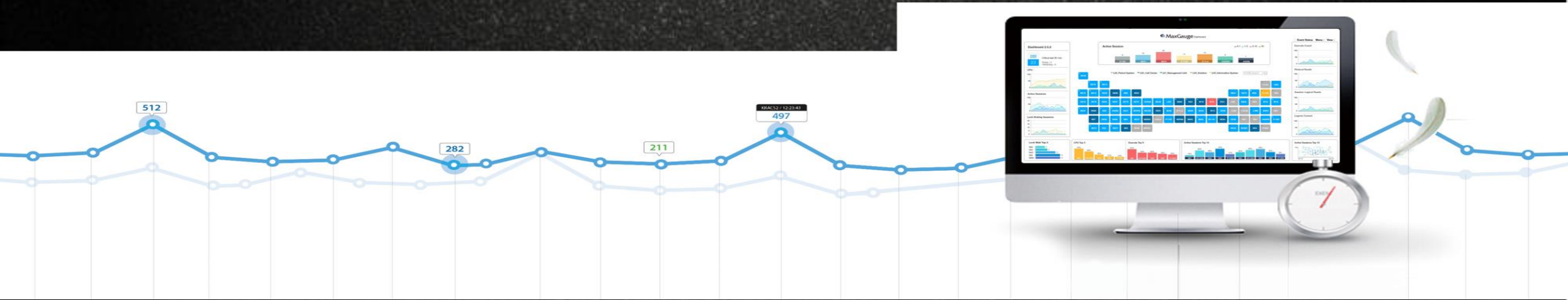
RTM & PA Main Feature

- Real-Time Monitor
- Performance Analyzer



Real-Time Monitor

1. 主要功能





RTM > 主要功能

"最優於資料庫性能管理的形態"



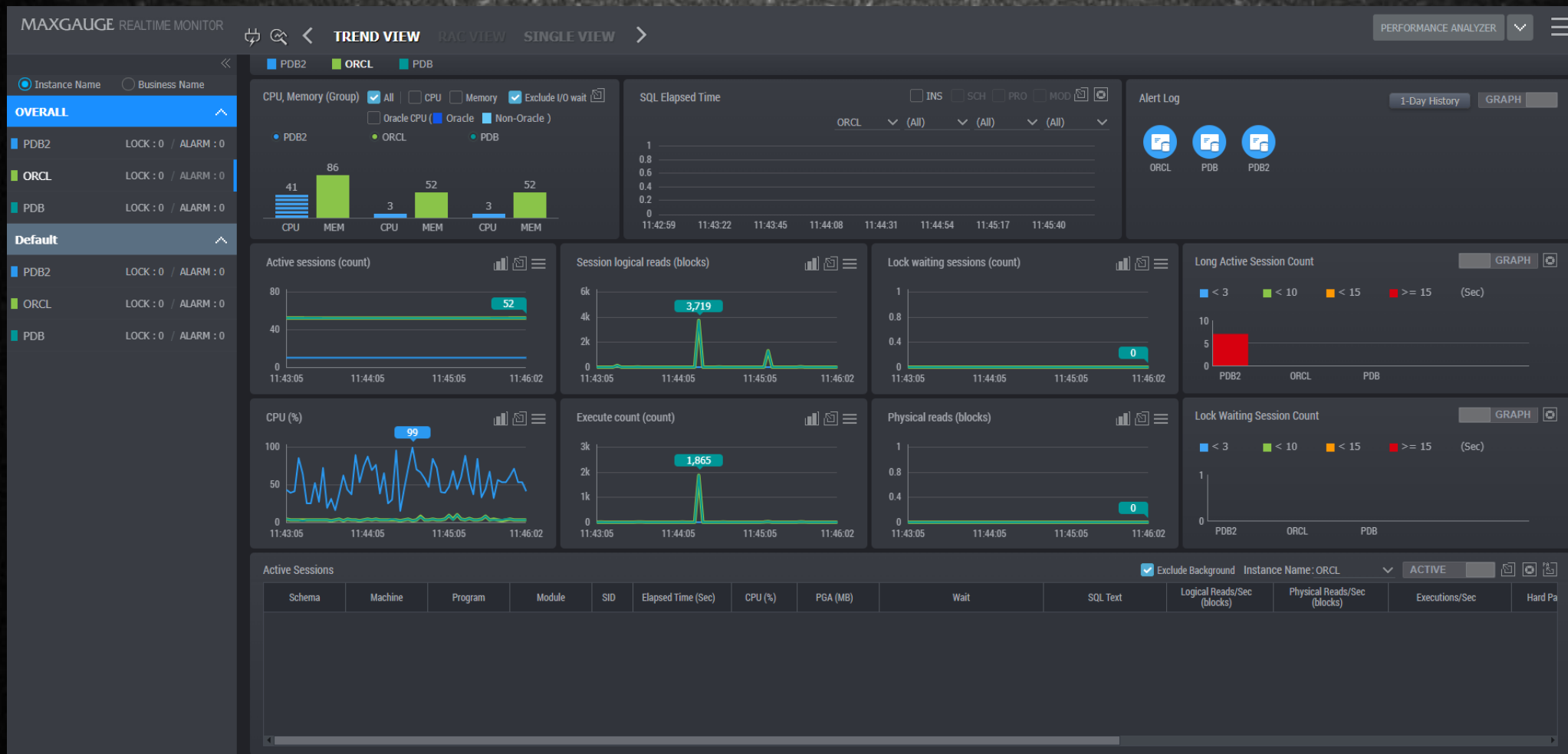
- 最優於同時監控多個資料庫的Trend View
- 添加專用框架可有效監控RAC的RAC View
- 便於對一個資料庫集中監控的Single View





RTM > Monitoring: Trend View

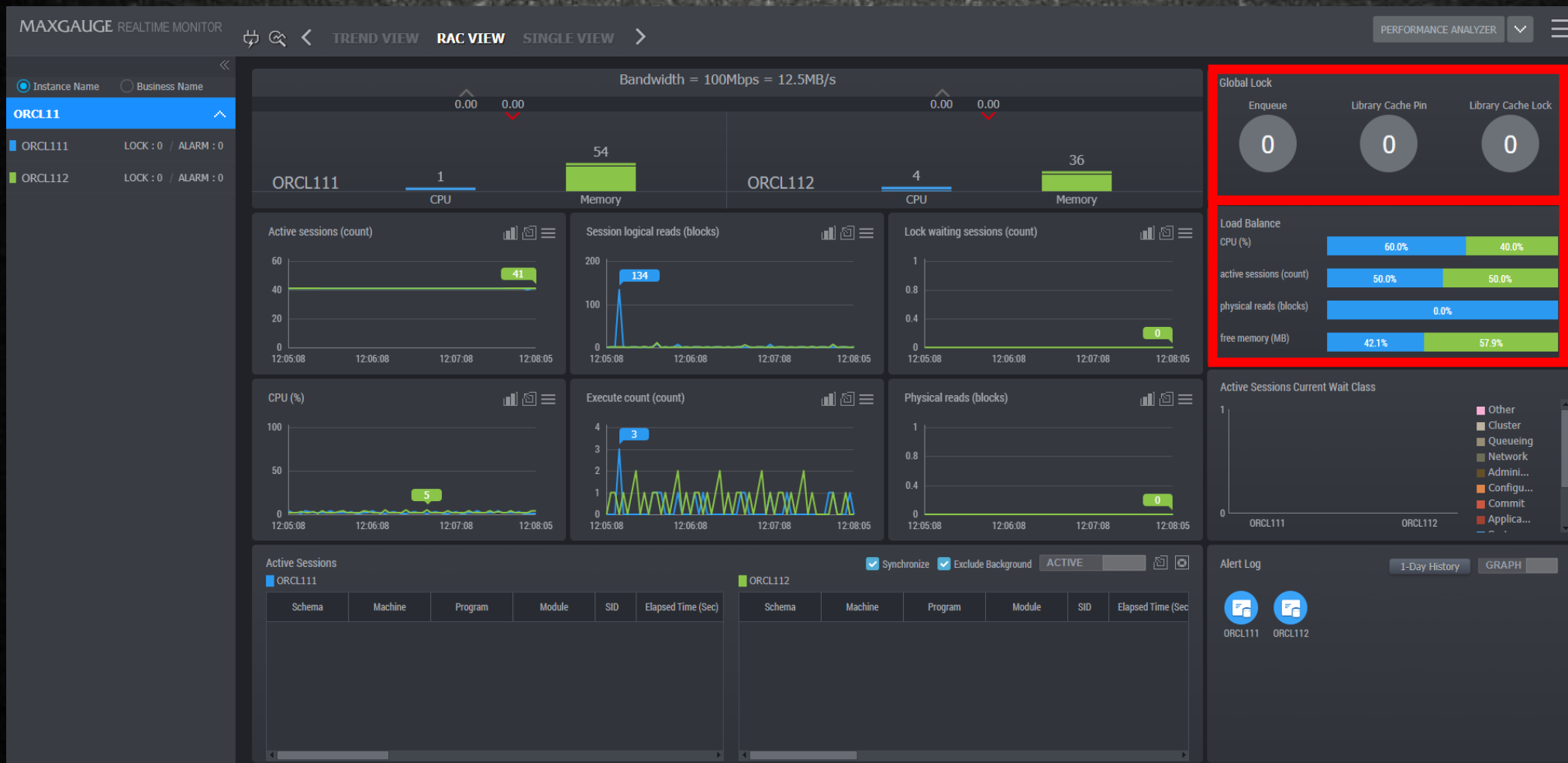
“可以同時監控多個資料庫的最優化的Trend View”





RTM > Monitoring: RAC View

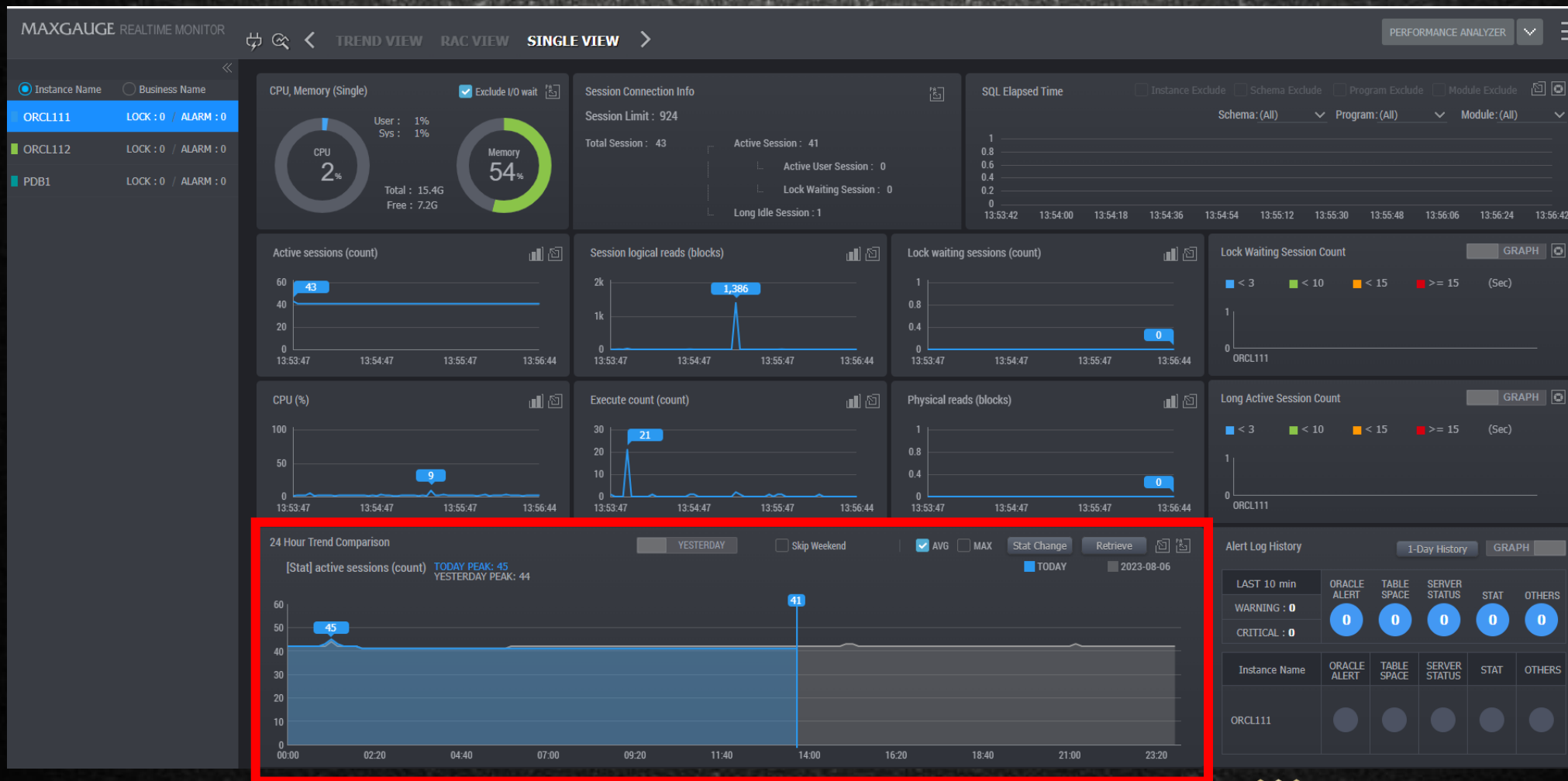
“引入專業化框架進行有效的RAC監控的RAC View”





RTM > Monitoring: Single View

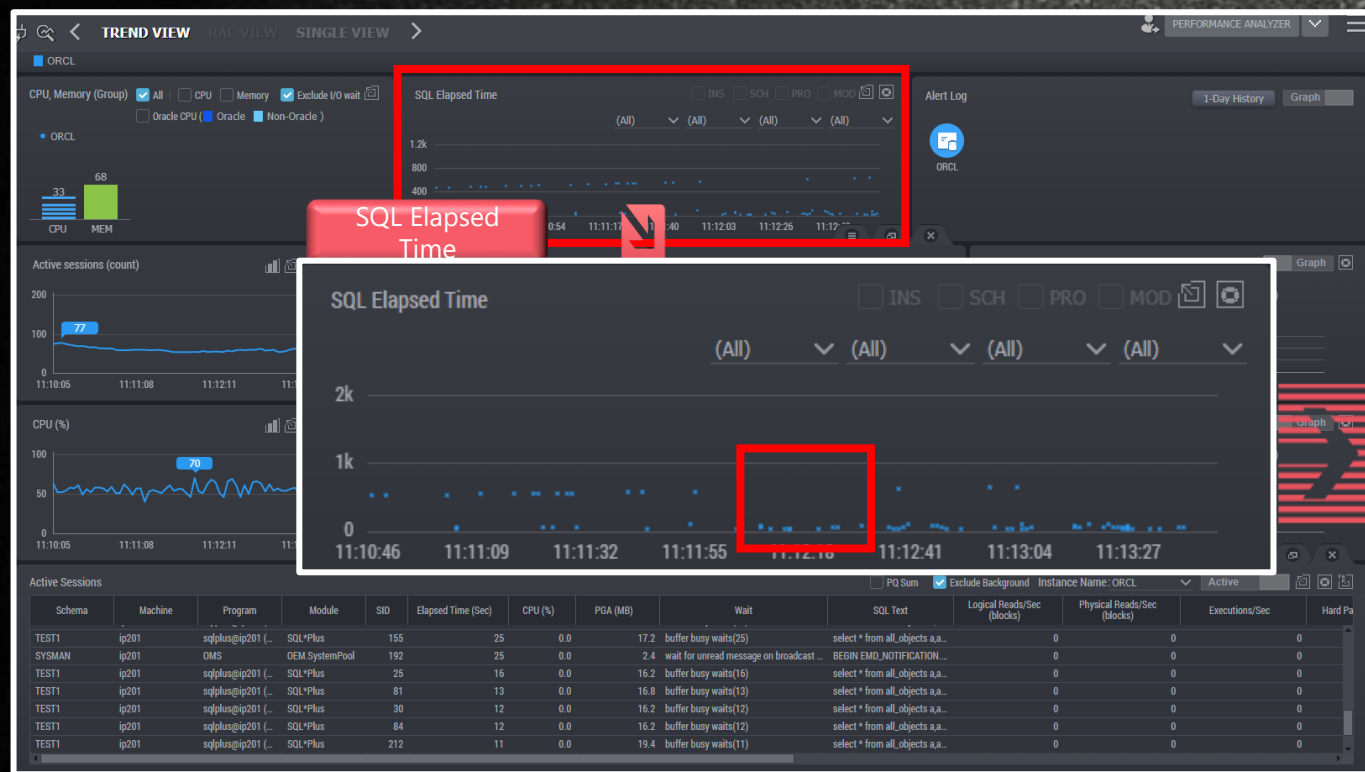
“有效對單一資料庫進行集中監控的Single View”





RTM > Monitoring: SQL Elapsed Time

“SQL Elapsed Time: SQL執行時間的即時診斷”



SQL Elapsed Time將SQL的執行時間以點單位顯示，以此可以確認SQL執行結束時間。顯示時，按實例以不同的顏色顯示，將拖選的區間以清單顯示，進一步可以確認SQL及SQL Detail信息。

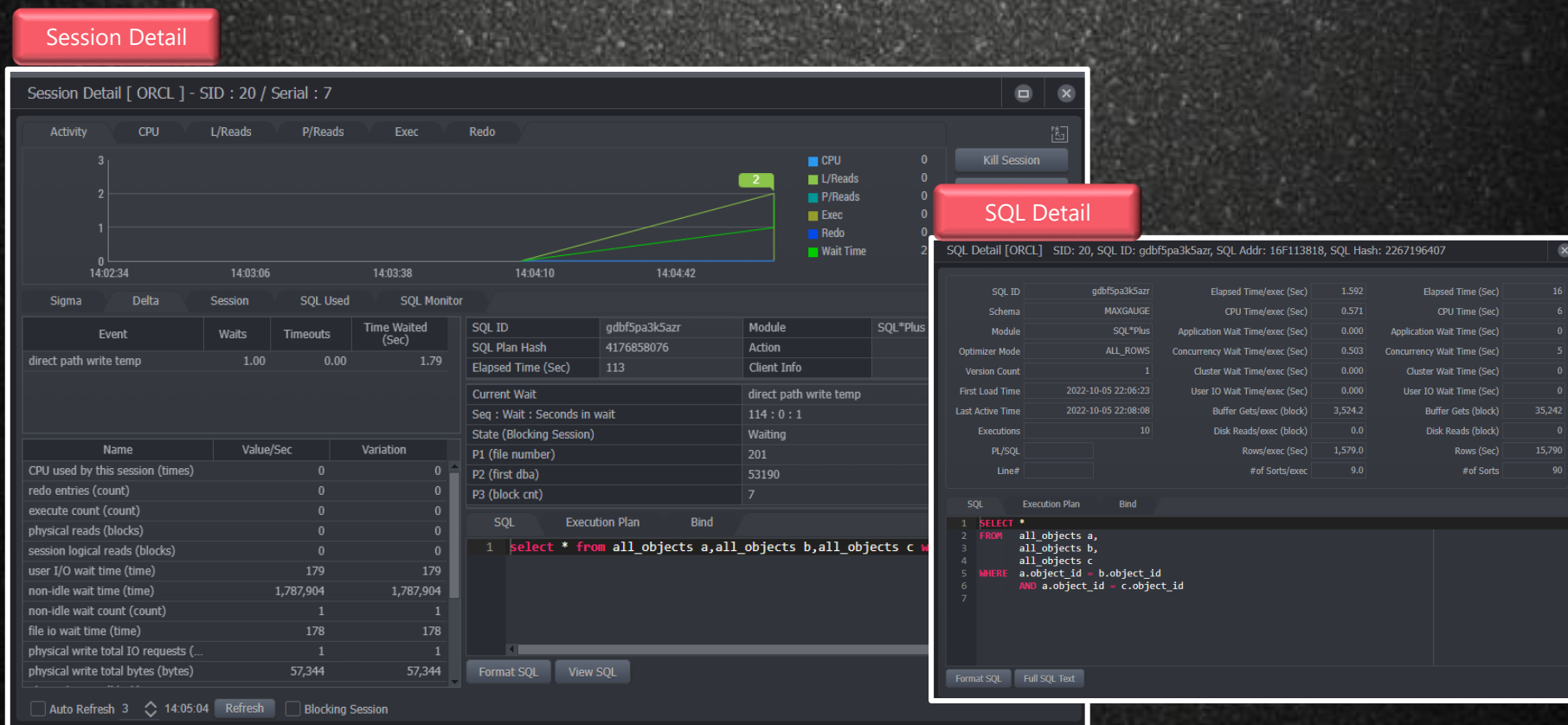


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RTM > Monitoring: Session Detail

“Session Detail: 所選會話的所有性能管理資料”





RTM > Monitoring: Customized Alarm Service

MAXGAUGE REALTIME MONITOR

TREND VIEW RAC VIEW SINGLE VIEW

Instance Name Business

OVERALL

ORCL **RTS Daer**

Default

ORCL LOCK : 0

SMS Schedule Editor

SMS Schedule Name: test

Weekly Schedule Setup

Clear

	SUN	MON	TUE	WED	THU	FRI	SAT
00							
01							
02							
03							
04							
05							
06							
07							
08							
09							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							

SMS Schedule Enable Period

☒ Always
☐ Specific Times

Add Delete

From	To	Time Window
------	----	-------------

SMS Schedule Disable Period

☒ Always
☐ Specific Times

Add Delete

From	To	Time Window
------	----	-------------

Save

1-Day History Graph

>= 15 (Sec)

Graph

>= 15 (Sec)

Graph

Active

Executions/Sec Hard Pa

0

0

CLOSE ALL X

1] Gather Disconnect

SYSMAN ip201 OMS OEM SystemPool 192 26 0.0 2.4 wait for unread message on broadcast ... BEGIN EMD_NOTIFICATION...





RTM > Monitoring: Session Locking/Blocking

MAXGAUGE REALTIME MONITORING

Session Detail [ORCL] - SID : 219 / Serial : 9

Activity CPU L/Reads P/Reads Exec Redo

1 0 14:30:49 14:31:11 14:31:33 14:31:55 14:32:17

0 0

CPU 0 L/Reads 0 P/Reads 0 Exec 0 Redo 0 Wait Time 0

Kill Session
Disconnect Session
Trace On
Trace Off

Sigma Delta Session SQL Used SQL Monitor

Event	Waits	Timeouts	Time Waited (Sec)
Current Wait			
Seq : Wait : Seconds in wait			
State (Blocking Session)			
P1 (driver id)			
P2 (#bytes)			

Name	Value/Sec	Variation
CPU used by this session (times)	0	0
redo entries (count)	0	0
execute count (count)	0	0
physical reads (blocks)	0	0
session logical reads (blocks)	0	0

Prev SQL ID	Module
8vqms8daszr70	sqlplus@ip201 (TNS V1-V3)
Prev SQL Plan Hash	Action
2226894183	
Elapsed Time (Sec)	Client Info
211	

Current Wait: SQL*Net message from client
Seq : Wait : Seconds in wait: 23 : 0 : 211
State (Blocking Session): Waiting
P1 (driver id): 1650815232
P2 (#bytes): 1

SQL Execution Plan Bind

1 update store set ID=3

Format SQL View SQL

☐ Auto Refresh 3 ☒ 14:32:31 Refresh ☐ Blocking Session

Alert Log

ORCL

29:31 14:30:28

29:31 14:30:28





RTM > Monitoring: Script Manager

Script Manager

Add Delete Delete All Modify Retrieve

Instance List

- Default
 - ☒ RAC11G1
 - ☒ RAC11G2
 - ☐ ORCL
 - ☐ ORCL11
 - ☐ SPC

Invalid Objects

SGA Stat

PGA Stat

Unanalyzed Segment

Undo Usage per Session

Backup Processing Tablespace

Undo Usage Summary

ASM

- Space Used in ASM Disk Groups
- Disk Contained in Disk Groups
- List of ASM filed as well as its vol.
- Disk Performance Metrics

User Script

Script Manager

Instance Name Group name Sector Size Block Size Allocation Unit Size State

腳本管理

添加 删除 全部删除 修改 检索

实例列表

ORA01

TRANSPORT_LAG

APPLY_LAG

Customize Script

藉由自定的script，
可對DG環境進行apply
狀態監控

Script Manager

Add Delete Delete All Modify Retrieve

Instance List

- ☐ JOB01
- ☒ ORA01

Instance Name	UPDATE_TIME	SERVICE_NAME	STATUS	HOSTNAME
ORA01	20161207210000	ora.oracleb3.AS...	ONLINE	oracleb3
ORA01	20161207210000	ora.oracleb3.LIS...	ONLINE	oracleb3
ORA01	20161207210000	ora.oracleb3.gsd	ONLINE	oracleb3
ORA01	20161207210000	ora.oracleb3.ons	ONLINE	oracleb3
ORA01	20161207210000	ora.oracleb3.vip	ONLINE	oracleb3
ORA01	20161207210000	ora.oracleb4.AS...	ONLINE	oracleb4
ORA01	20161207210000	ora.oracleb4.LIS...	ONLINE	oracleb4
ORA01	20161207210000	ora.oracleb4.gsd	ONLINE	oracleb4
ORA01	20161207210000	ora.oracleb4.ons	ONLINE	oracleb4
ORA01	20161207210000	ora.oracleb4.vip	ONLINE	oracleb4
ORA01	20161207210000	ora.oradb.db	OFFLINE	oracleb3
ORA01	20161207210000	ora.oradb.oracleb1...	OFFLINE	oracleb3
ORA01	20161207210000	ora.oradb.oracleb2...	ONLINE	oracleb4
ORA01	20161207210000	ora.oradb.oracleb...	ONLINE	oracleb4
ORA01	20161207210000	ora.oradb.oracleb...	ONLINE	oracleb3
ORA01	20161207205900	ora.oradb.oracleb...	ONLINE	oracleb4

Customize Script

藉由自定的script，
可對CRS狀態監控

ASM

- Space Used in ASM Disk Groups
- Disk Contained in Disk Groups
- List of ASM filed as well as its vol.
- Disk Performance Metrics

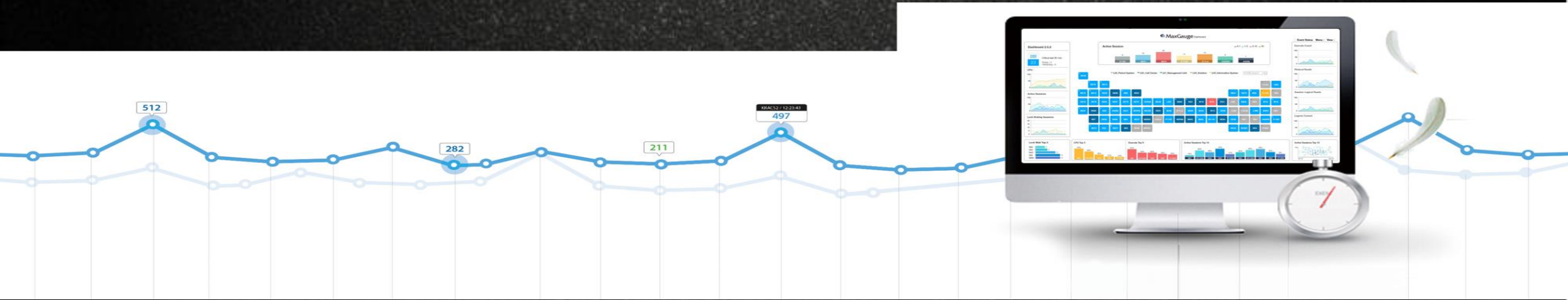
User Script

CRS_CHECK (admin)

Auto Refresh 3 16:44:28 Refresh

Performance Analyzer

1. 主要功能





PA > 主要功能

提供數種資料庫分析方法



- 為了精密的性能及障礙分析的**Performance Analysis**
- 適用資料庫視覺化技術的分析畫面**Data Visualization**
- 提供從多種觀點進行性能比較的**Power Comparison**
- 記錄資料庫容量趨勢的**Capacity Planning**
- 輕鬆比較SQL執行計畫變更歷史的**SQL Plan Analysis**





PA > Analyzing: Visualizing

“直觀分析的 **Data Visualization**”

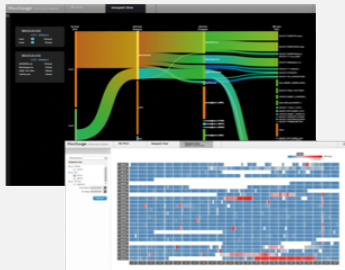
• DIAGNOSIS

• POST-ANALYSIS

• SQL TUNING



REAL-TIME MONITORING



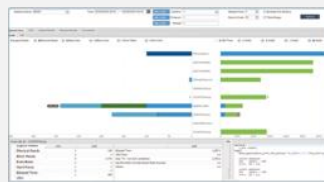
PERFORMANCE ANALYSIS



STAT/EVENT/RATIO



TOP SESSION



TOP SQL



SESSION DETAILS



WAIT EVENT DETAILS



SQL DETAILS



SQL TUNING

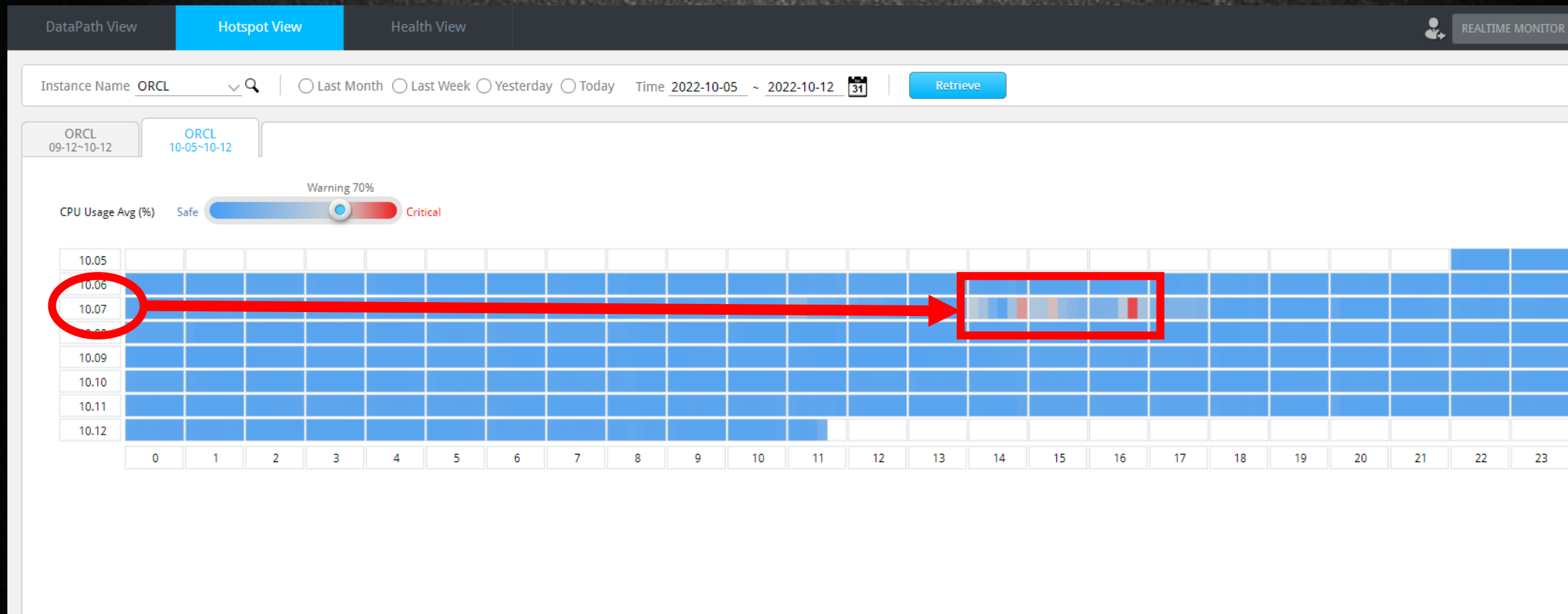


EXECUTION PLAN ANALYSIS



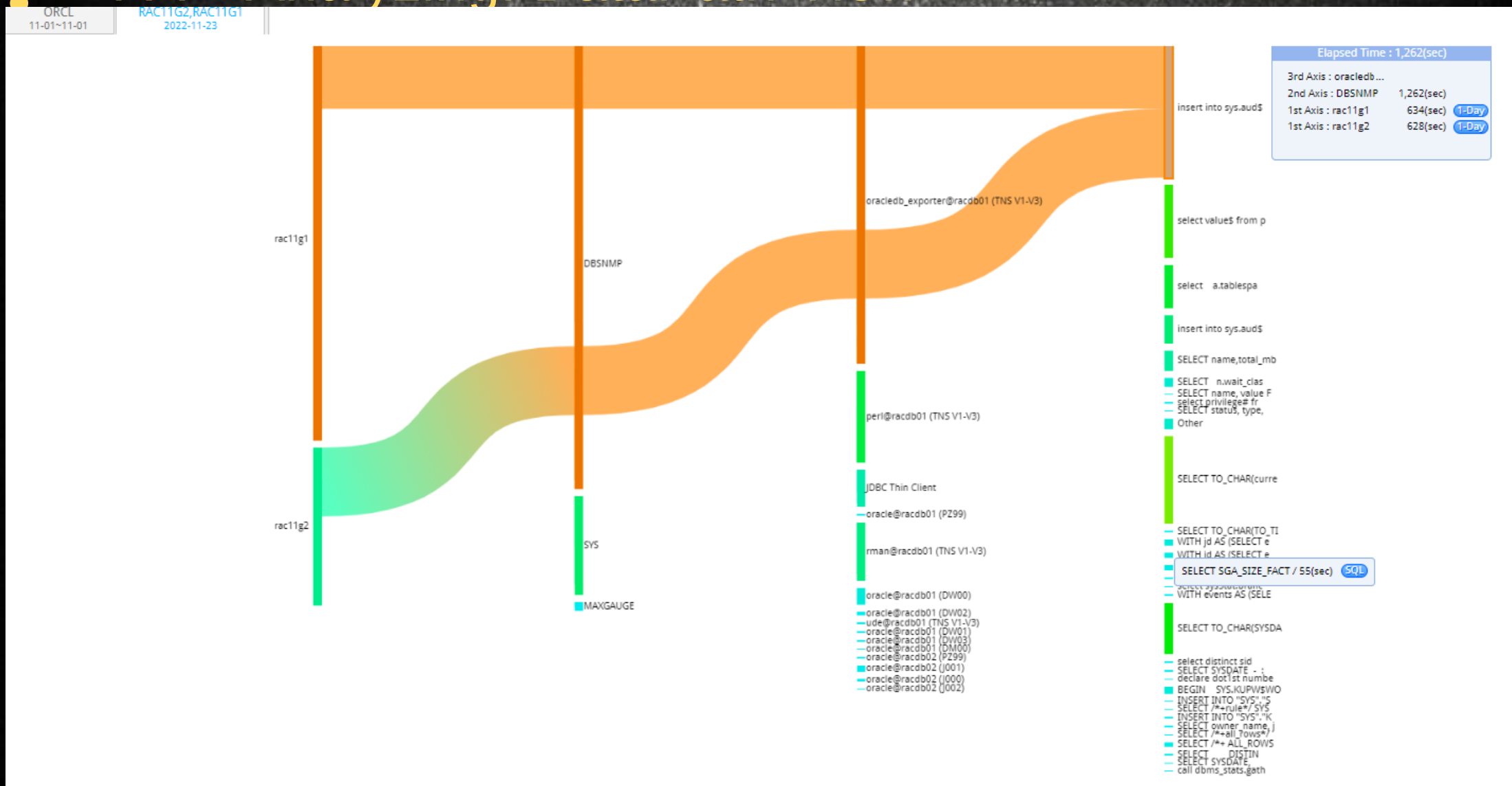


PA > Analyzing: Hotspot View





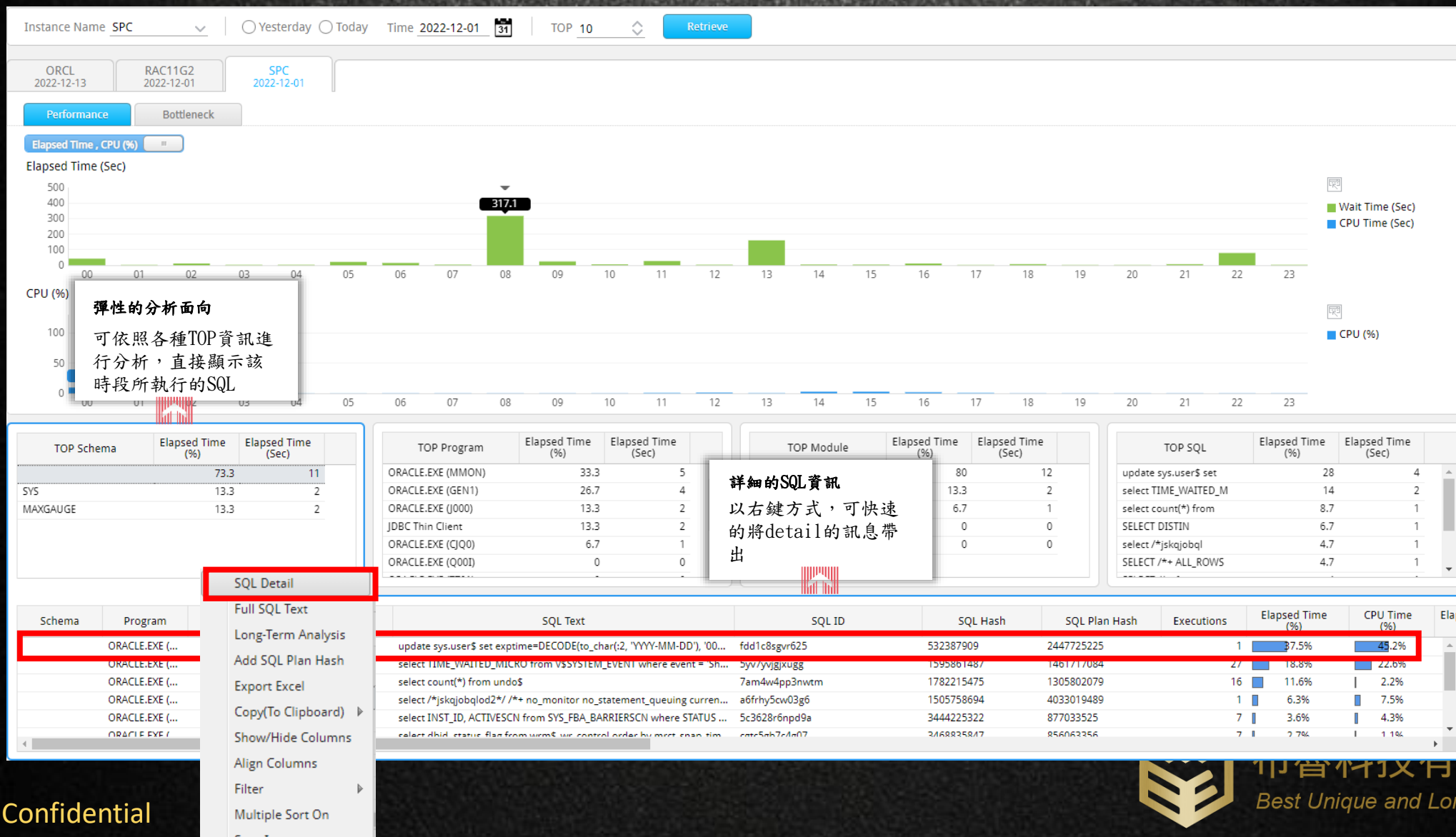
PA > Analyzing: DataPath View





PA > Analyzing: 1-Day Summary View

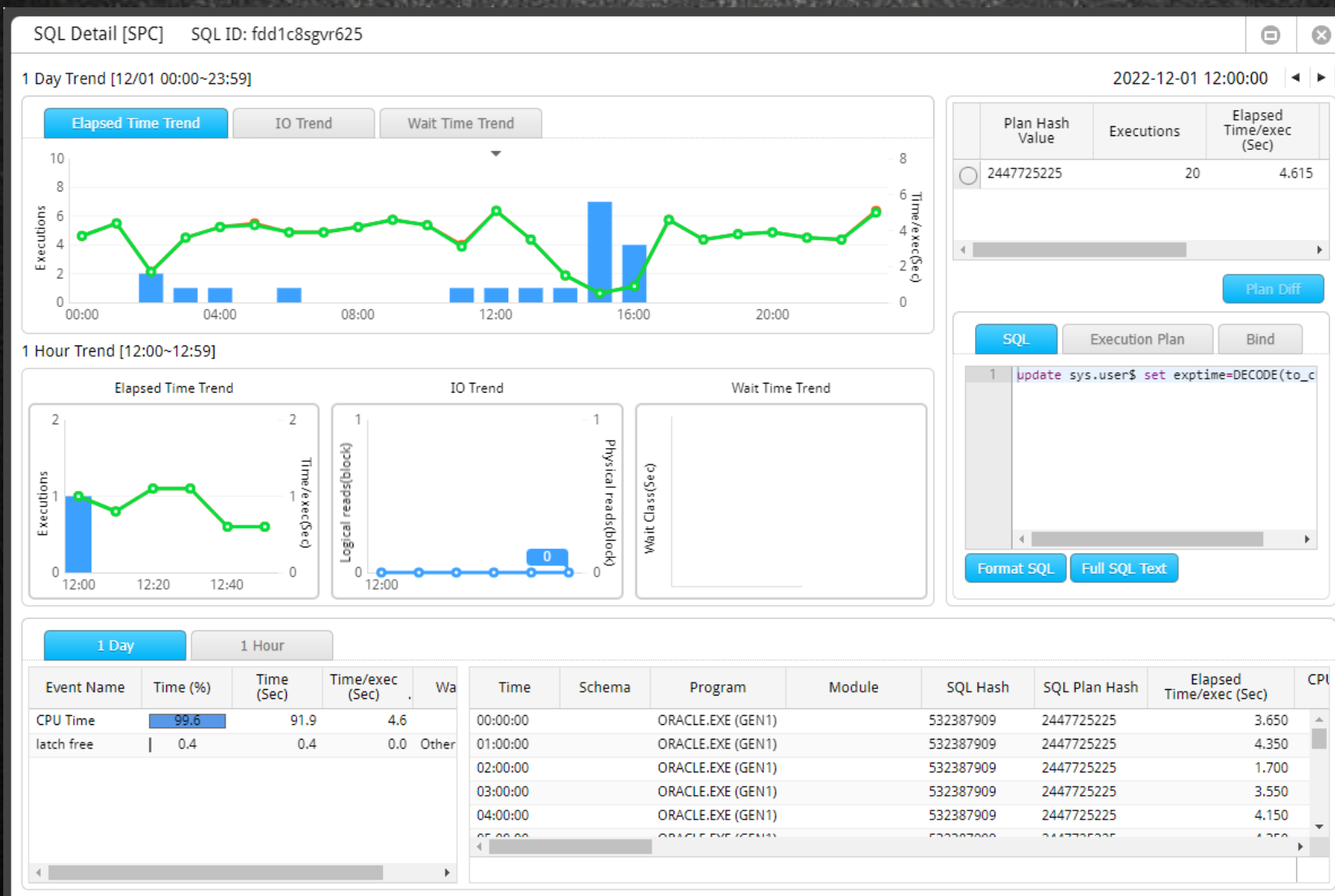
“輕鬆快速的看出當天每個時段的負載狀況”





PA > Analyzing: SQL Detail

“輕鬆快速的看出該SQL執行狀況”



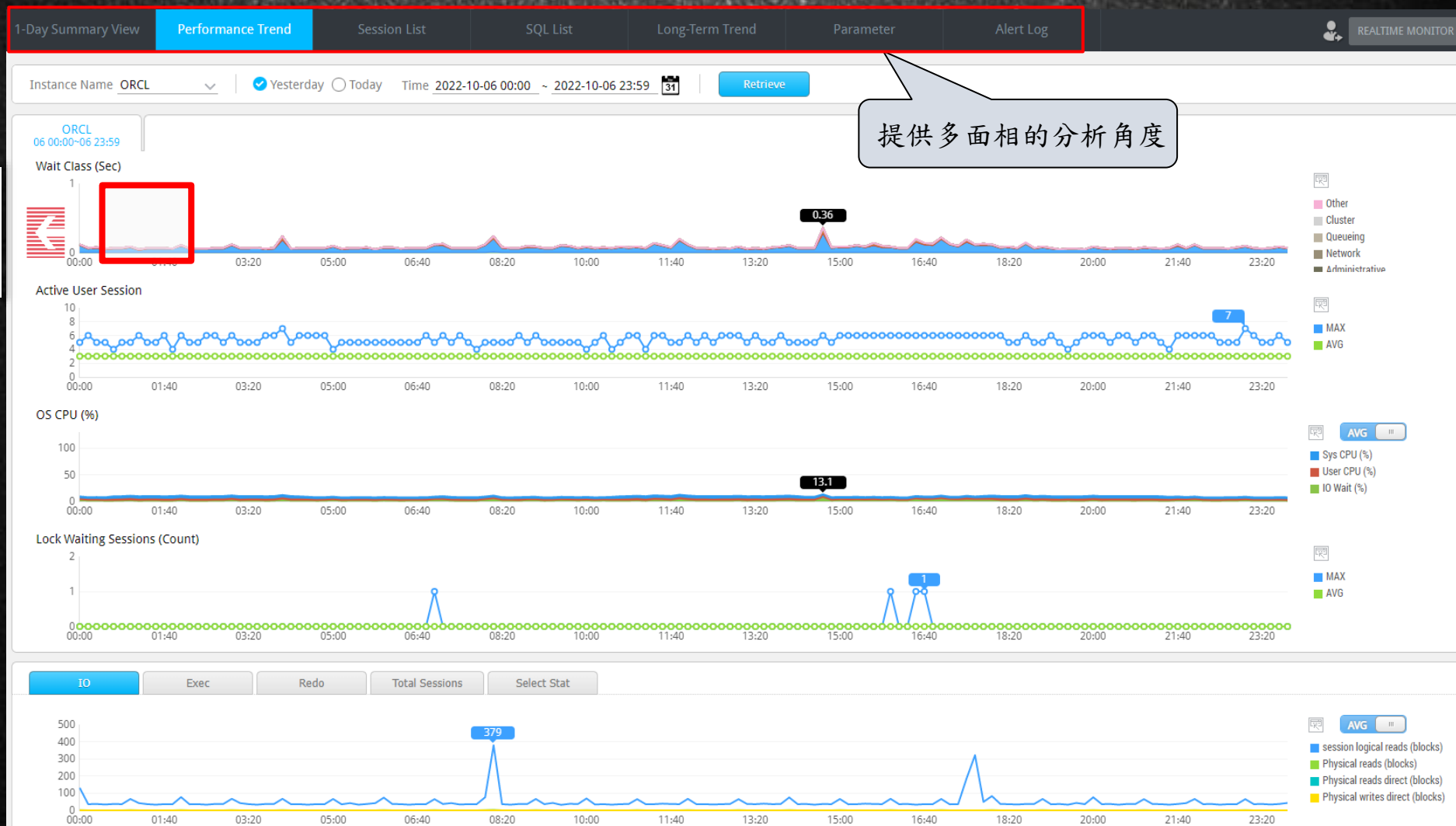


PA > Analyzing: Performance Trend

“彈性多面向的分析方法”

直觀式的操作方式

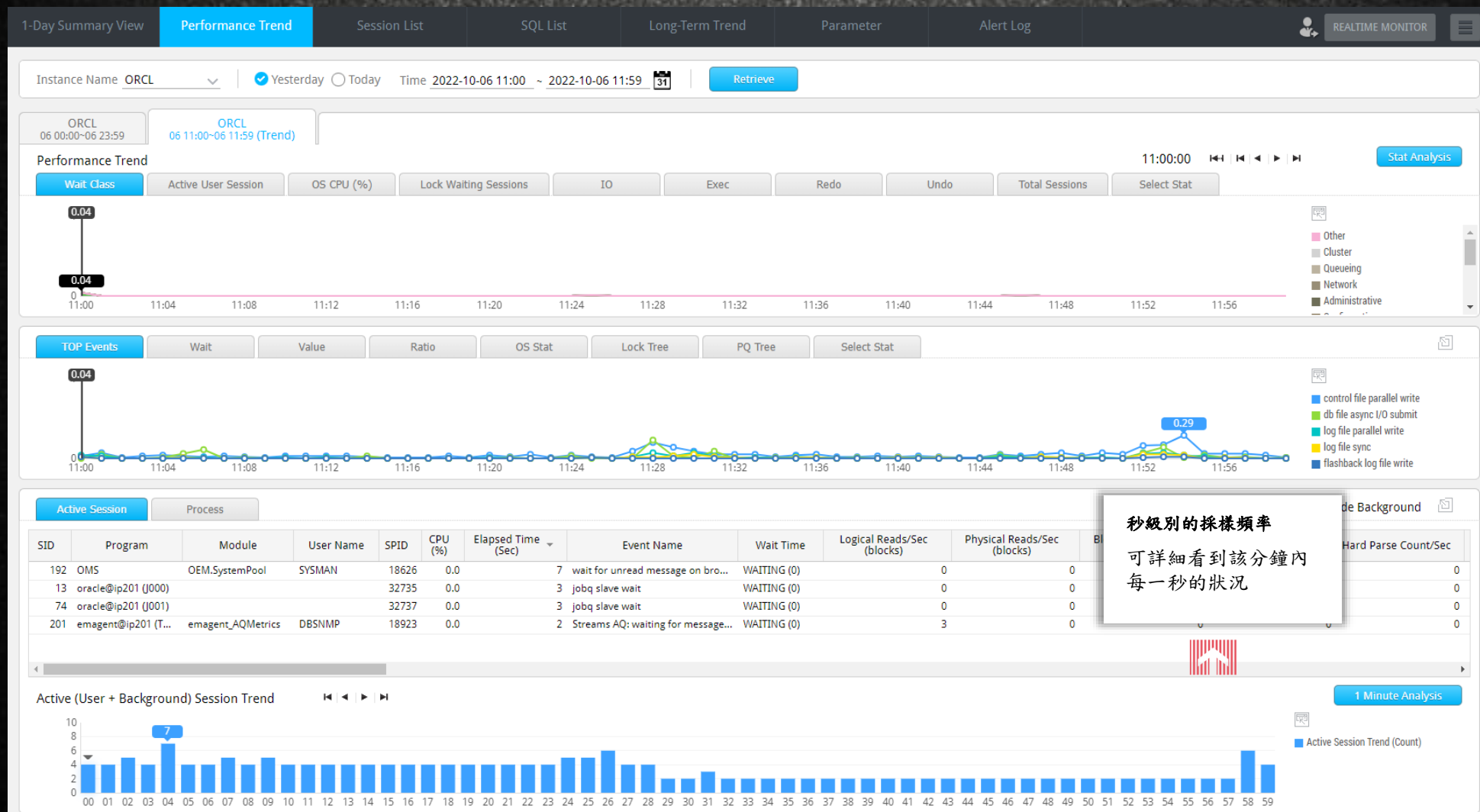
以框選的方式選擇想要查詢的區段





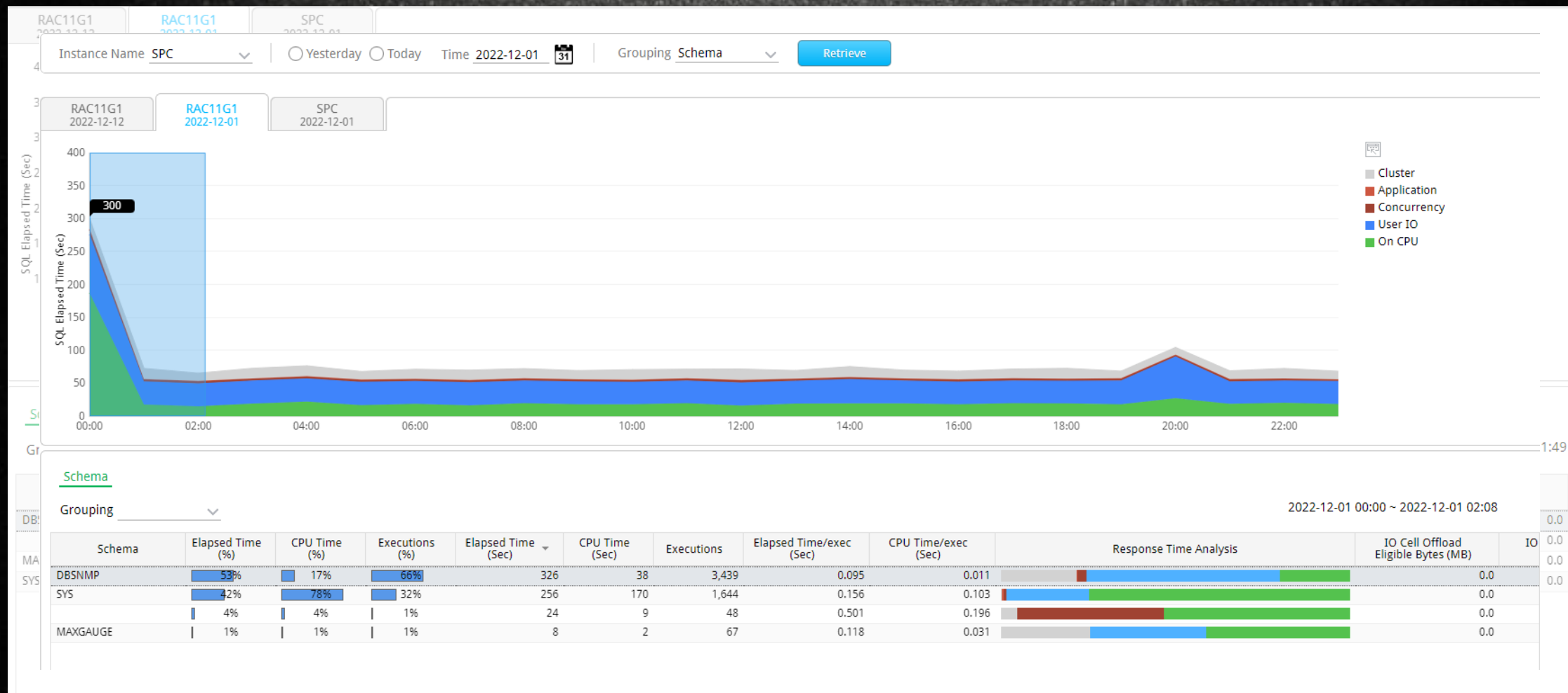
PA > Analyzing: Performance Trend

“秒級別的歷程記錄”



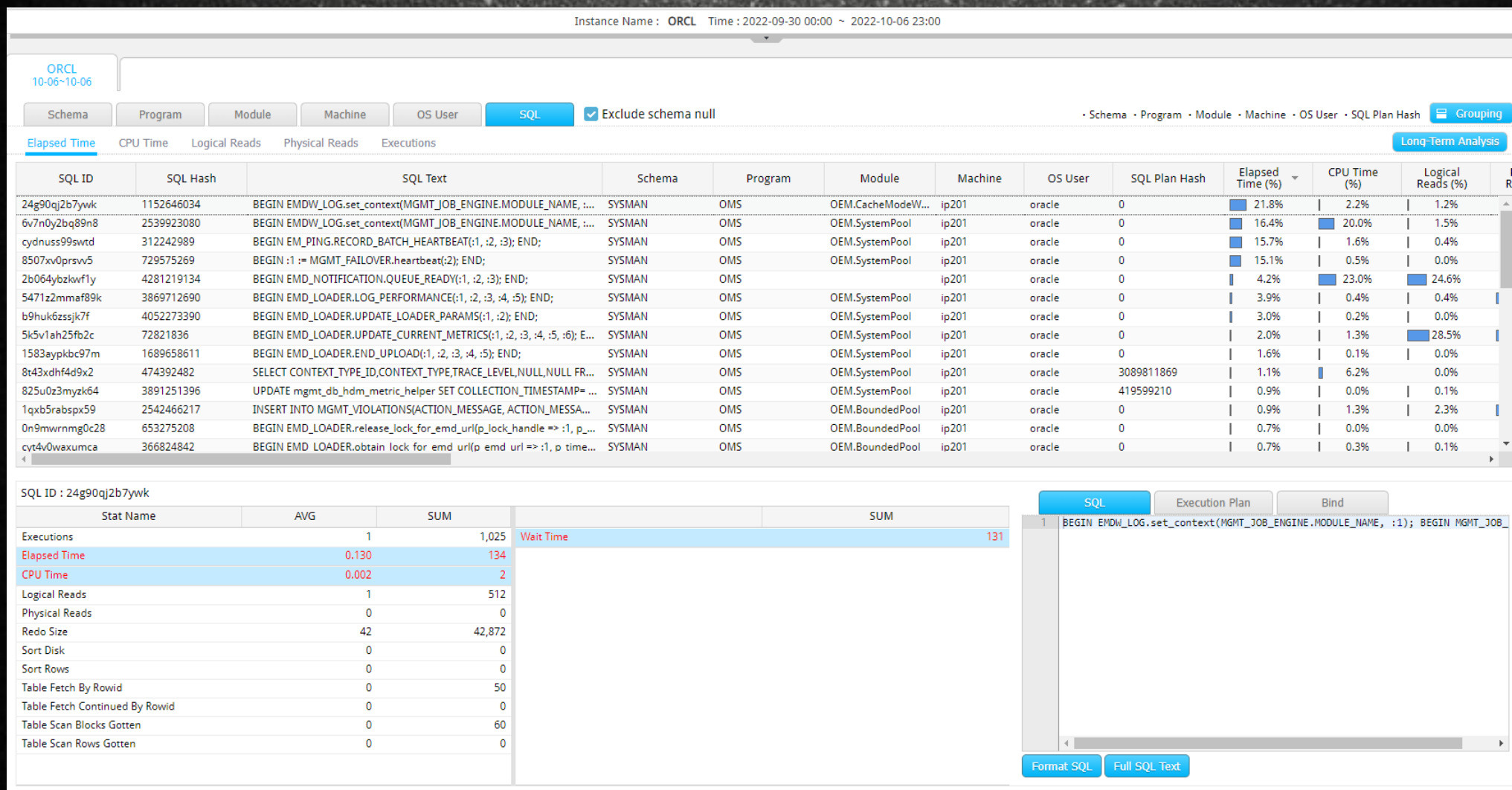


PA > Analyzing: Elapsed Time Analysis





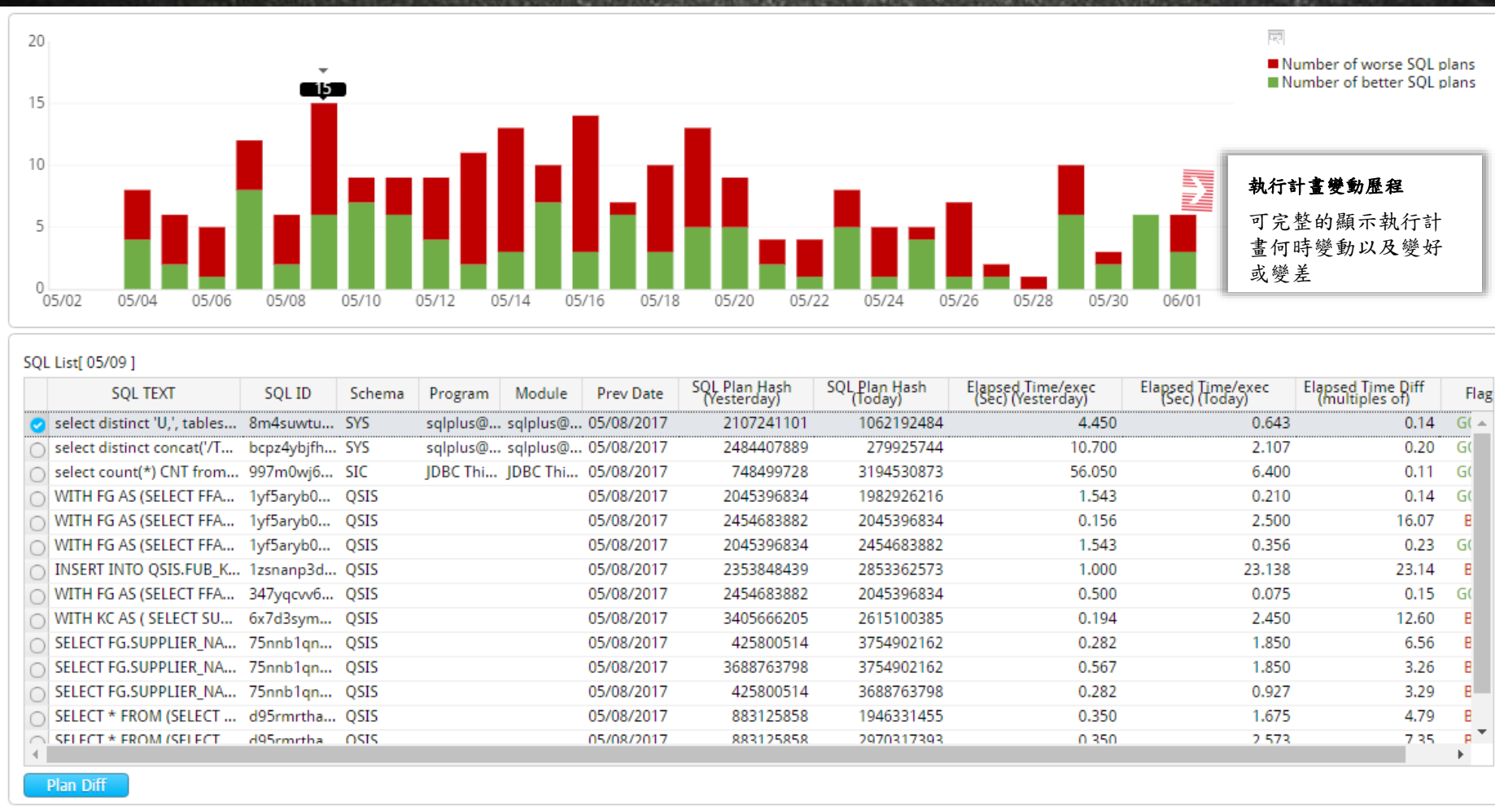
PA > Analyzing: Top-N Analysis





PA > Analyzing: SQL Plan Analysis

“執行計畫變動的分析”





“快速的對異動前後的執行計畫進行比較”

SQL Plan Diff SQLID: 997m0wj6g2g9x



PA > Analyzing: Parameter Manage

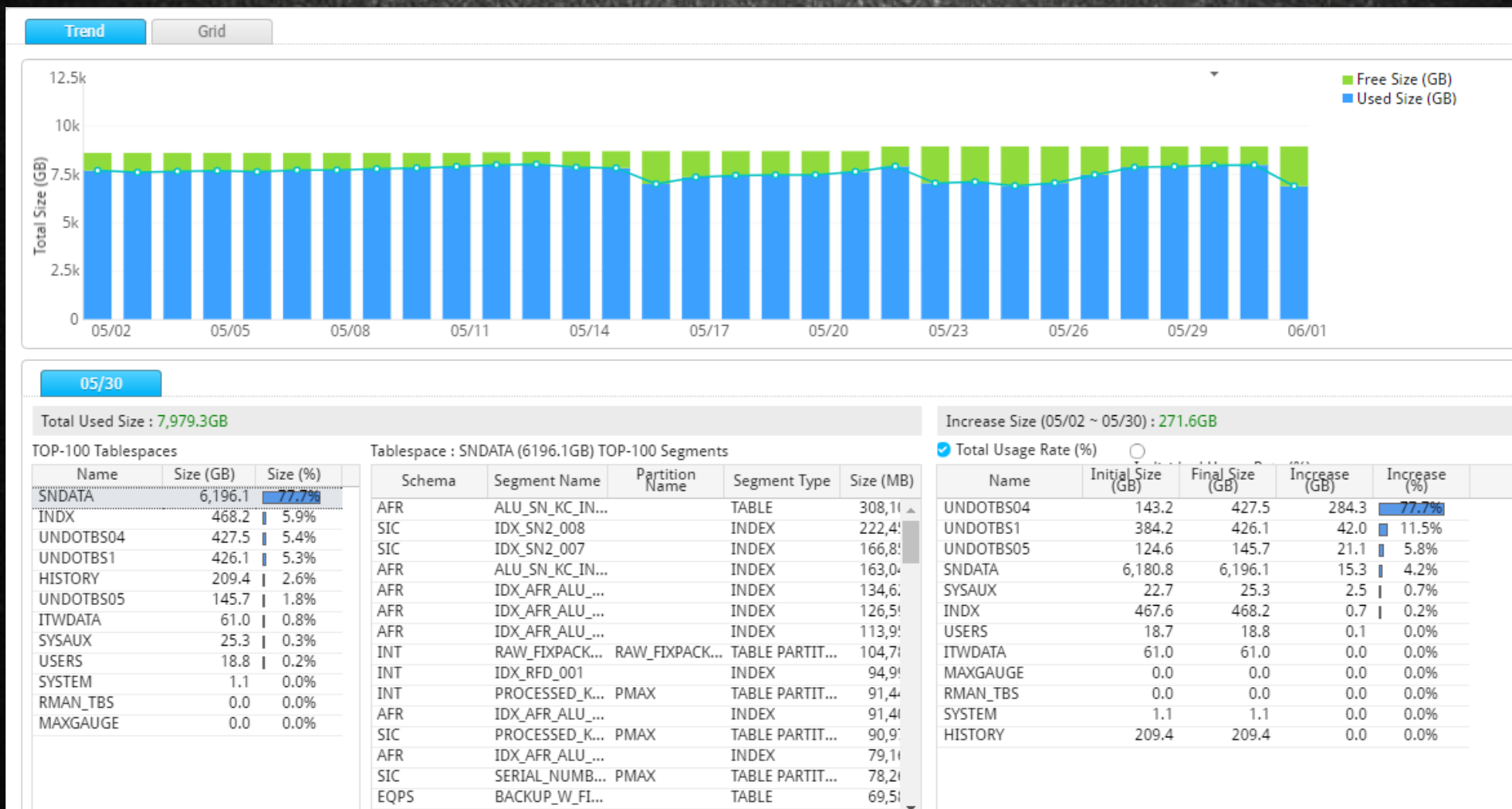
Pre-Check					
Search Result					
Modification History					
Name	Date	Current Value	Prev Value		
▾ _cursor_db_buffers_pi...	2016-03-24	214	70		
▾ _db_block_buffers	2016-03-24	32406	10802		
▾ _db_block_hash_buckets	2016-03-24	65536	32768		
▾ _db_block_hash_latches	2016-03-24	2048	1024		
▾ __db_cache_size	2016-03-29	146800640	197132288		
	2016-03-28	197132288	226492416		
	2016-03-27	226492416	234881024		
	2016-03-24	260046848	92274688		
▾ _db_file_exec_read_co...	2016-03-24	128	43		
▾ db_file_multiblock_rea...	2016-03-24	128	43		
▾ _db_flash_cache_keep...	2016-03-24	201715792	205747792		
▾ _db_writer_coalesce_a...	2016-03-24	2752512	917504		
▾ _fastpin_enable	2016-03-24	201792769	205824257		
▾ memory_max_target	2016-03-24	0	415236096		
▾ memory_target	2016-03-24	0	415236096		
▾ __pga_aggregate_target	2016-03-24	12582912	142606336		
▾ pga_aggregate_target	2016-03-24	10485760	0		
trace_enabled	TRUE	TRUE	FALSE	IMMEDIATE	FALSE enable in memory tracing





PA > Analyzing: DB Capacity

“快速的得知資料成長趨勢”



THE END

Q & A

